Next week quiz.

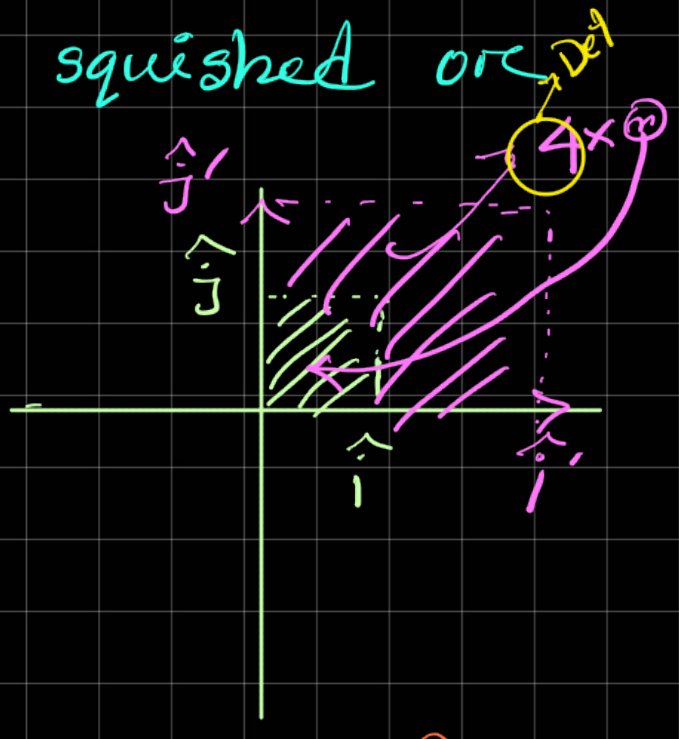
# Determinants

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & -7 & 12 \\ 3 & 1 & 5 \end{bmatrix}$$

$$\det(A) = |A| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & -7 & 12 \\ 3 & 1 & 5 \end{vmatrix}$$

how much the area is squished or stretched.

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$




H.W

Why det of only sq matrix (n×n)?

## 10 properties of a determinant:

① Det of identity Matrix is 1 →
cz it doesn't scale anything

$\det(I) = 1$

② $\det(P) = 1$ or $-1$ → even # of permutation
↳ odd #

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

P ↳ $-1$

not scaling but $\hat{i}$ & $\hat{j}$ flipped.

V.I.

③ $$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$$

↓ det of original + ↓ det of changed part

$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\begin{vmatrix} c & d \end{vmatrix} = \begin{vmatrix} c & d \end{vmatrix}$$

so det is a linear operation as I am scaling & adding (linear op)

$$\begin{vmatrix} ta & tb \\ tc & td \end{vmatrix} = t \begin{vmatrix} a & b \\ tc & td \end{vmatrix}$$

$$= t^2 \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$\det(A+B) = \det(A) + \det(B)$ → Not correct

↳ exercise

Is row wise op. being held or not?

④ $$\begin{vmatrix} a & b & c \\ a & b & c \\ d & e & f \end{vmatrix} = 0$$

$$\begin{vmatrix} a & b & c \\ a & b & c \\ d & e & f \end{vmatrix} = x$$ any 2 rows changed, it would be $-x$.

(cz $\det(P) = 1$ or $-1$.

$$\begin{vmatrix} a & b & c \\ a & b & c \\ d & e & f \end{vmatrix} = -x$$

for which number $x = -x$? → 0.

⑤ $$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c-al & d-bl \end{vmatrix}$$

$r_2 = r_2 + (l) \times r_1$

$$= \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a & b \\ -al & -bl \end{vmatrix}$$

$$= \begin{vmatrix} a & b \\ c & d \end{vmatrix} + -l \begin{vmatrix} a & b \\ a & b \end{vmatrix} \quad \begin{vmatrix} a & b \\ a & b \end{vmatrix} = 0$$

$$= \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$ → same as original.

row operations do not change the determinant / col space

↳ so scaling remains same.

⑥ Row of zeroes. → Det = 0

$$\begin{vmatrix} 0 & 0 \\ c & d \end{vmatrix} = 0 \begin{vmatrix} ? & ? \\ c & d \end{vmatrix}$$

$0 \times$ smth.

$= 0.$

⑦

$$\begin{vmatrix} p_1 & \times & \times \\ 0 & p_2 & \times \\ 0 & 0 & p_3 \end{vmatrix}$$

↓ upper triangular

everything under & above pivots are 0.

reduce

$$\begin{vmatrix} p_1 & 0 & 0 \\ 0 & p_2 & 0 \\ 0 & 0 & p_3 \end{vmatrix}$$

did some elimination.

$$= p_1 \begin{vmatrix} 1 & 0 & 0 \\ 0 & p_2 & 0 \end{vmatrix} = p_1 p_2 p_3 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix}$$

$$\begin{vmatrix} 0 & 0 & p_3 \end{vmatrix} \qquad \begin{vmatrix} 0 & 0 & 1 \end{vmatrix}$$

$$= p_1 p_2 p_3 \times 1$$

For upper triangular Mat, the det is the product of pivots.

⑧ $\det(A) = 0$, then $A$ is singular (Non invertible)

At least one pivot should be zero.

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 2 \begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{vmatrix} = 2 \cdot 0 = 0$$

⑨ $\det(A \cdot B) = \det(A) \cdot \det(B)$

3 | ↓2

1st B

then A

1st squish or stretch rowspace by the det of B. then A.

$$\det(A^{-1}) = ? \frac{1}{\det(A)}$$

$$A^{-1}A = I$$

$$\det(AA^{-1}) = \det(I)$$

$$\therefore \det(A^{-1})\det(A) = 1$$

$$\therefore \det(A^{-1}) = \frac{1}{\det(A)}$$

(10) $\det(A^T) = \det(A)$

as square mat, the row & col space of $A$ & $A^T$ are same.

$$A = LU$$

$$A^T = (LU)^T = U^T L^T$$

$|A|$

$= |LU|$

$= \underbrace{|L|}_{1} \underbrace{|U|}_{x}$

$|A^T|$

$= |U^T L^T|$

$= |U^T| |L^T|$ (upper become lower by Trans posing)

$x \quad 1$

pivots won't change (can't be 1)

$= P_1 P_2 P_3$

$= P_1 P_2 P_3$




## det of lower triangular Mat.

$$\begin{vmatrix} P_1 & 0 & 0 \\ x & P_2 & 0 \\ x & x & P_3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x & x & 1 \end{vmatrix}$$

$A = L\ U$

↓

will have 1 in pivots

cz U will have the real pivots.

both can't contain same pivots.

If 2 columns are same det 0.

cz transposing the rows could make the rows col. So 0 det.

(*) $\det(A+A)$ , $A_{(n\times n)}$

$= 2^n \det(A)$

$\det(2A)$

$= 2^3 \times \det(A)$ if $(3\times 3)$ $\begin{bmatrix} 2a & 2b & 2c \\ 2d & 2e & 2f \\ 2g & 2h & 2i \end{bmatrix}$

for $n\times n \rightarrow 2^n \times \det(A)$ $\quad A+A$

$$\Rightarrow 2^3 \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$= 2^3 \times \det|A|$

next cramer's rule.